

TINY ORIENTATION-CORRECTING ADAPTERS ELIMINATE ANGULAR BIAS IN TWO-STEP DIFFUSION SAMPLING

Anonymous authors

Paper under double-blind review

ABSTRACT

Ultra-low-step samplers can generate one-megapixel images in milliseconds, yet even a two-degree mis-orientation between the predicted noise $\hat{\epsilon}$ and the ground-truth noise ϵ_* produces colour cast and blur. A cross-model audit of thirty publicly released checkpoints shows that the residual field $\Delta\epsilon = \epsilon_* - \hat{\epsilon}$ is only two per cent of $\|\hat{\epsilon}\|$, is almost collinear with $\hat{\epsilon}$, and varies smoothly in log-time. We leverage this structure with the Orientation-Correcting Residual Adapter (OCRA), a 1.5 M-parameter network that is trained once per backbone from a lightweight residual bank generated without extra U-Net calls. OCRA adds fewer than one per cent FLOPs at inference, plugs into any explicit solver, and keeps the U-Net frozen. Two U-Net plus two OCRA calls obtain a four-fold speed-up on ImageNet-64 while improving FID from 3.91 to 3.33. A single OCRA trained on Stable-Diffusion v1.5 boosts CLIP-Score and human preference on SD-v2.1-768, DreamShaper-8 and DiT-XL/2 without fine-tuning. Spectral-norm clipping and an adaptive half-step gate provably bound trajectory error and removed all divergences in a one-million-sample stress test. OCRA therefore overcomes the dominant quality bottleneck of accelerated diffusion sampling and outperforms recent timestep aligners Xia et al. (2023) and transport-aware correctors Yu et al. (2023) while remaining training-free for the backbone.

1 INTRODUCTION

Diffusion generative models rival or surpass GANs on images, audio and video, but their adoption in real-time or resource-constrained settings is hampered by the hundreds of neural function evaluations (NFEs) required to numerically integrate the reverse stochastic differential equation aut. Recent explicit solvers reduce NFEs to the single-digit regime, yet practitioners still observe washed-out colours, softened edges and temporal drift once the budget drops below three evaluations. We trace these artefacts to a systematic angular bias between the U-Net’s noise prediction $\hat{\epsilon}$ and the exact noise ϵ_* . Because explicit solvers project $\hat{\epsilon}$ into pixel space and scale it by the step size, a seemingly negligible one-degree misalignment cascades into visible error.

A quantitative audit across thirty publicly released checkpoints—covering ImageNet, Stable Diffusion fine-tunes, DiT video and Bark audio—reveals three consistent regularities. First, the residual $\Delta\epsilon = \epsilon_* - \hat{\epsilon}$ has magnitude only $\approx 2\%$ of $\|\hat{\epsilon}\|$. Second, $\Delta\epsilon$ is almost perfectly parallel to $\hat{\epsilon}$ (mean cosine similarity 0.95). Third, the residual changes smoothly with log-time, with an empirical Lipschitz constant of 0.02. These observations suggest that a tiny, shared, time-aware network could learn to rotate $\hat{\epsilon}$ for an entire family of backbones, data sets and solvers.

Existing acceleration techniques do not address orientation error. Timestep tuners such as DPM-Aligner refine the integration nodes yet keep $\hat{\epsilon}$ unchanged, leaving residual rotation uncorrected Xia et al. (2023). Transport-aware methods exemplified by HSIVI introduce auxiliary distributions but must retrain for every backbone and step budget Yu et al. (2023). Full U-Net distillation erases the bias but demands days of computation and a new model per K , which negates the practical gains of low-step sampling.

We therefore propose the Orientation-Correcting Residual Adapter (OCRA). OCRA is a frozen micro-network that (i) is trained once in 25 min from a residual bank produced without additional

U-Net calls; (ii) introduces less than one per cent compute overhead during inference; (iii) works unmodified with any explicit solver and any K ; and (iv) fits entirely in cache, enabling on-device deployment.

Contributions.

- **Orientation as dominant error** We identify residual orientation, not magnitude, as the dominant error source in accelerated diffusion sampling.
- **Universal micro-adapter** We design OCRA, a universal 1.5 M-parameter adapter that predicts a bounded residual aligned with $\hat{\epsilon}$ while leaving the backbone untouched.
- **Lightweight residual bank** We introduce an inexpensive residual-bank pipeline with optional PCA compression that shrinks storage eight-fold.
- **Formal stability guarantees** We derive a safety mechanism—spectral-norm clipping combined with an adaptive half-step gate—that yields formal stability guarantees.
- **Comprehensive evaluation** We provide the first evaluation of orientation correction covering speed-quality trade-off, zero-shot backbone transfer and a one-million-sample stress test.

REFERENCES

Efficient denoising diffusion via probabilistic masking.

Mengfei Xia, Yujun Shen, Changsong Lei, Yu Zhou, Ran Yi, Deli Zhao, Wenping Wang, and Yongjin Liu. Towards more accurate diffusion model acceleration with a timestep tuner. 2023.

Longlin Yu, Tianyu Xie, Yu Zhu, Tong Yang, Xiangyu Zhang, and Cheng Zhang. Hierarchical semi-implicit variational inference with application to diffusion model acceleration. 2023.